

We work with the **time-shifted Fréchet likelihood**

$$\log \mathcal{L}^{X_t, LDM}(\theta; A, \alpha) = \sum_{t=1}^T \log f(x_t - \theta t),$$

where for the Fréchet distribution:

$$f(x) = A\alpha(Ax)^{-(1+\alpha)} \exp(-(Ax)^{-\alpha}), \quad x > 0.$$

1. Log-likelihood expression

Let $y_t = x_t - \theta t$, and $z_t = Ay_t$.

$$\ell(\theta, A, \alpha) = T \log(A\alpha) - (1 + \alpha) \sum_{t=1}^T \log(Ay_t) - \sum_{t=1}^T (Ay_t)^{-\alpha}$$

✓ 2. First-order derivatives

◆ w.r.t. θ

$$\frac{\partial \ell}{\partial \theta} = (1 + \alpha) \sum_{t=1}^T \frac{t}{y_t} - \alpha A \sum_{t=1}^T t (Ay_t)^{-\alpha-1}$$

◆ w.r.t. A

$$\frac{\partial \ell}{\partial A} = -\frac{\alpha T}{A} + \alpha \sum_{t=1}^T y_t (Ay_t)^{-\alpha-1}$$

◆ w.r.t. α

$$\frac{\partial \ell}{\partial \alpha} = \frac{T}{\alpha} - \sum_{t=1}^T \log(Ay_t) + \sum_{t=1}^T (Ay_t)^{-\alpha} \log(Ay_t)$$

✓ 3. Second-order derivatives

◆ $\theta\theta$

$$\frac{\partial^2 \ell}{\partial \theta^2} = +(1 + \alpha) \sum_{t=1}^T \frac{t^2}{y_t^2} - \alpha(\alpha + 1) A^2 \sum_{t=1}^T t^2 (A y_t)^{-\alpha-2}$$

◆ AA

$$\frac{\partial^2 \ell}{\partial A^2} = \frac{\alpha T}{A^2} - \alpha(\alpha + 1) \sum_{t=1}^T y_t^2 (A y_t)^{-\alpha-2}$$

◆ $\alpha\alpha$

$$\frac{\partial^2 \ell}{\partial \alpha^2} = -\frac{T}{\alpha^2} - \sum_{t=1}^T (A y_t)^{-\alpha} [\log(A y_t)]^2$$

◆ Cross derivatives

θ - A

$$\frac{\partial^2 \ell}{\partial \theta \partial A} = -\alpha \sum_{t=1}^T t (Ay_t)^{-\alpha-1} + \alpha(\alpha+1)A \sum_{t=1}^T t y_t (Ay_t)^{-\alpha-2}$$

θ - α

$$\frac{\partial^2 \ell}{\partial \theta \partial \alpha} = \sum_{t=1}^T \frac{t}{y_t} - A \sum_{t=1}^T t (Ay_t)^{-\alpha-1} + \alpha A \sum_{t=1}^T t (Ay_t)^{-\alpha-1} \log(Ay_t)$$

A - α

$$\frac{\partial^2 \ell}{\partial A \partial \alpha} = -\frac{T}{A} + \sum_{t=1}^T y_t (Ay_t)^{-\alpha-1} [1 - \alpha \log(Ay_t)]$$